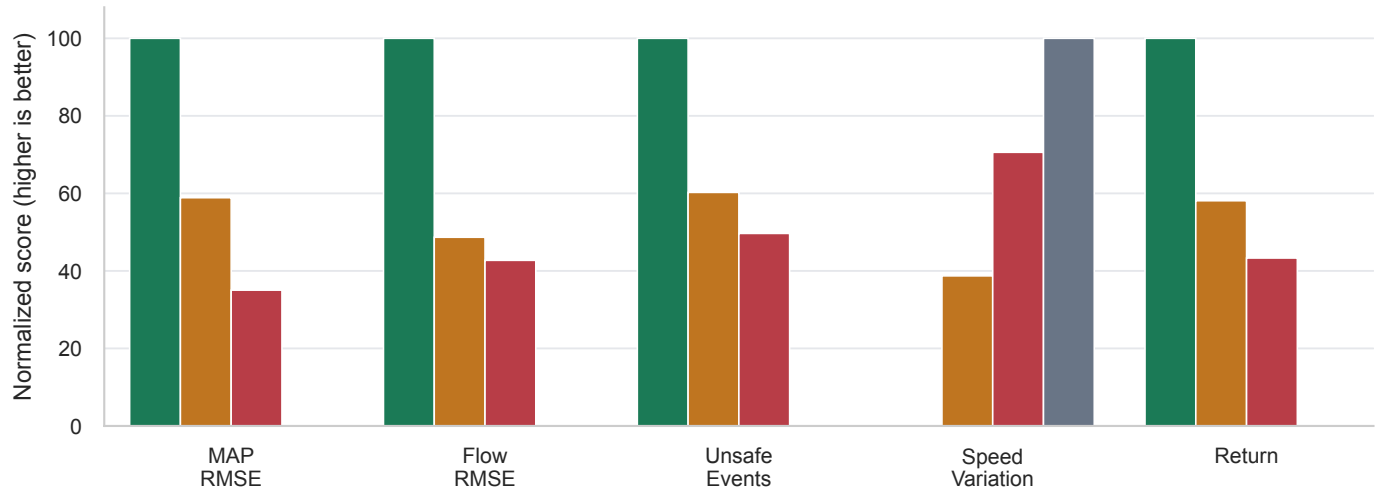


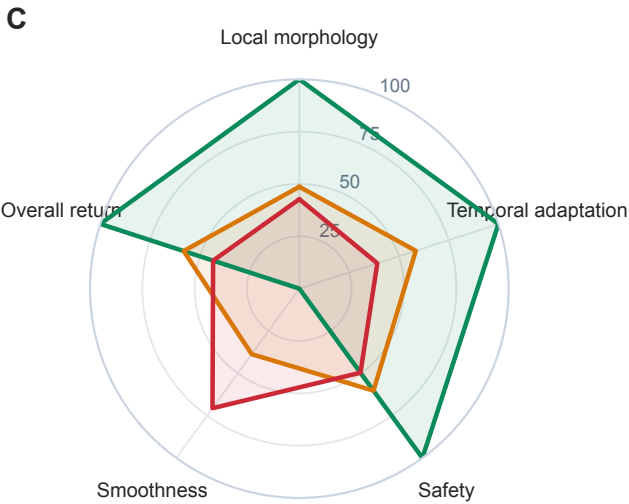
Ablation study of CNN and Transformer modules

Synthetic pilot data generated from repeated virtual runs; replace the CSV with measured results for final reporting.

A Normalized control quality across ablations



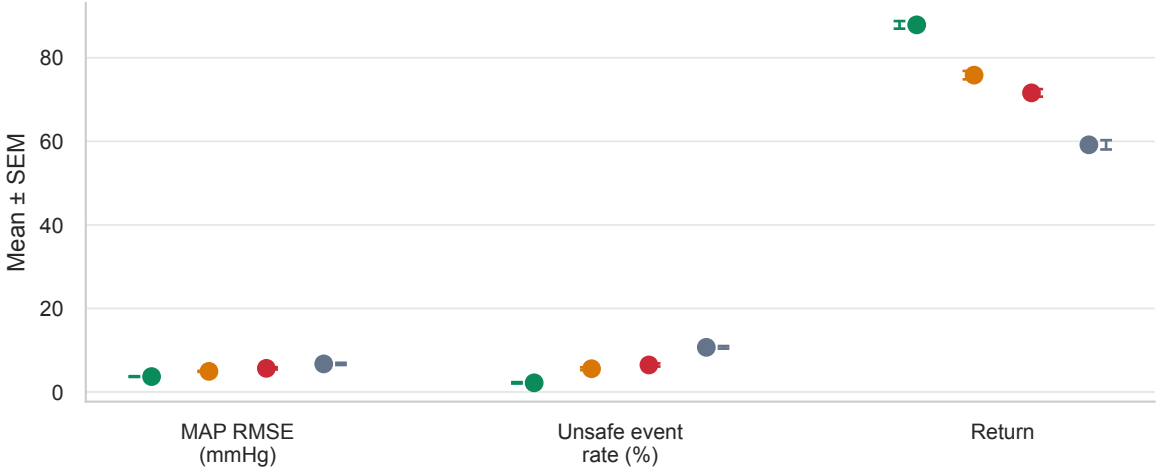
C Module-level contribution profile



B Performance trade-off relative to full model



D Representative raw metrics



Full CNN-Transformer w/o CNN w/o Transformer MLP/RNN Baseline

Error bars indicate SEM across virtual runs. Error-type metrics are inverted before normalization in panel A.